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Asymmetric Catalysis |*Hot Paper*|

Palladium-Catalyzed Enantioselective Alkoxycarbonylation of Propargylic Carbonates with (*R*)- or (*S*)-3,4,5-$(MeO)_3$-MeOBIPHEP

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Abstract: (*R*)- and (*S*)-3,4,5-$(MeO)_3$-MeOBIPHEP have been identified as the efficient chiral ligands for the palladium-catalyzed highly enantioselective synthesis of 2,3-allenoates from different types of easily available racemic propargylic carbonates with 90–98% *ee* and decent yields. The potential of the products was demonstrated with high chirality transfer efficiency.

Allenes are a class of compounds characterized by two cumulated carbon–carbon double bonds exhibiting extraordinary activity as building blocks or even as chiral ligands in organic synthesis.[1,2] Their unique structure has also been found in a great number of natural products and pharmaceutical molecules, which has led to ample interest in the synthesis of allenes.[1a,3] Recently, many methodologies for the synthesis of allenes have been developed;[4] however, controlling the axial chirality in trisubstituted allenes, especially the trisubstituted 2,3-allenoates, which are extremely useful precursors for the synthesis of other allenes due to the versatile reactivity of the ester functionality, remains a challenging task.[5] A common method to generate such optically active compounds is through the kinetic resolution of racemic compounds (Scheme 1 a):[6] Chiral trisubstituted allenoates were recovered from the 1,3-dipolar cycloaddition of racemic 2,3-allenoates using a bisphosphoric acid catalyst, resulting in 35–48% yield and 85–99% *ee*, together with the products, 3-methylenepyrrolidine derivatives in 39–57% yield and 64–94% *ee*. Another efficient method to access enantiomerically enriched trisubstituted allenoates relies on the dynamic kinetic resolution-based isomerization of 1-substituted 3-alkynoates, from which the starting 3-alkynoates is recovered even after 24 h, but is barely possible to separate from 2,3-allenoates (Scheme 1 b).[7] An alternate approach utilizes organozinc reagents as the nucleophile through the corresponding S_N2' substitution with optically active propargylic mesylates (Scheme 1 c).[8] With the help of DMSO, 2,3-allenoates could be afforded with 87% yield and 98% *ee*, whereas only one example of a trisubstituted allenoate was demonstrated; synthesis of optically active trisubstituted allenoates may also be realized through Pd-catalyzed alkoxycarbonylation of optically active propargylic mesylates in 69–90% yield and 80–99% *ee* (Scheme 1 d).[9] Asymmetric olefination of ketenes is another option albeit using at least a stoichiometric amount of chiral ylide (Scheme 1 e).[10]

Catalytic enaotioselective synthesis is the most ideal strategy in consideration of cost and atom economy. In the presence of a chiral phosphite ligand, chiral 2,3-allenoates can be synthesized enantioselectively through Pd-catalyzed asymmetric β-hydride elimination of the non-readily available stereodefined enol triflates (Scheme 1 f).[11] However, only 20% *ee* was detected in the synthesis of trisubstituted allenoates. Applying intermolecular addition of nitroalkanes and activated enynes to the synthesis of chiral trisubstituted allenoates is also feasible (Scheme 1 g).[12] High yields and excellent enatioselectivities were obtained in the tandem addition/isomerization process, although in some cases the alkynoate intermediates were detected, which can be further transformed into allenoates under the identical reaction conditions. Asymmetric C–H insertion of α-diazoesters into terminal alkynes is an alternate method to access chiral trisubstituted allenoates (Scheme 1 h).[13] Catalyzed by copper(I) complexes and chiral cationic guanidinium salts, enantiomerically enriched allenoates may be produced. Apart from these, chiral trisubstituted allenoates may be synthesized from the relatively complicated cyclopropenonylmethyl acetates catalyzed by a Lewis base with good yields and enatioselectivities (Scheme 1 i).[14]

In 2013, our laboratory disclosed a catalytic asymmetric synthesis of 2,3-allenoates with **ECNU-Phos** (see Table 1) starting from propargylic carbonates.[15] However, the enantioselectivity was not satisfactory in some cases. Here, we report our recent results in identifying (*R*)- or (*S*)-3,4,5-$(MeO)_3$-MeOBIPHEP (MeO-BIPHEP = 2,2′-bis(diphenylphosphino)-6,6′-dimethoxy-1,1′-biphenyl) as the ligand for a very general Pd-catalyzed highly enantioselective synthesis of chiral trisubstituted allenoates from cheap and readily available and differently substituted 2-alkynylic carbonates with about 90–98% *ee* (Scheme 2).

Racemic methyl 1-phenyl-2-heptyn-1-yl carbonate [(±)-**1a**] was used as the model compound with $[(\pi\text{-allyl})PdCl]_2$ as the

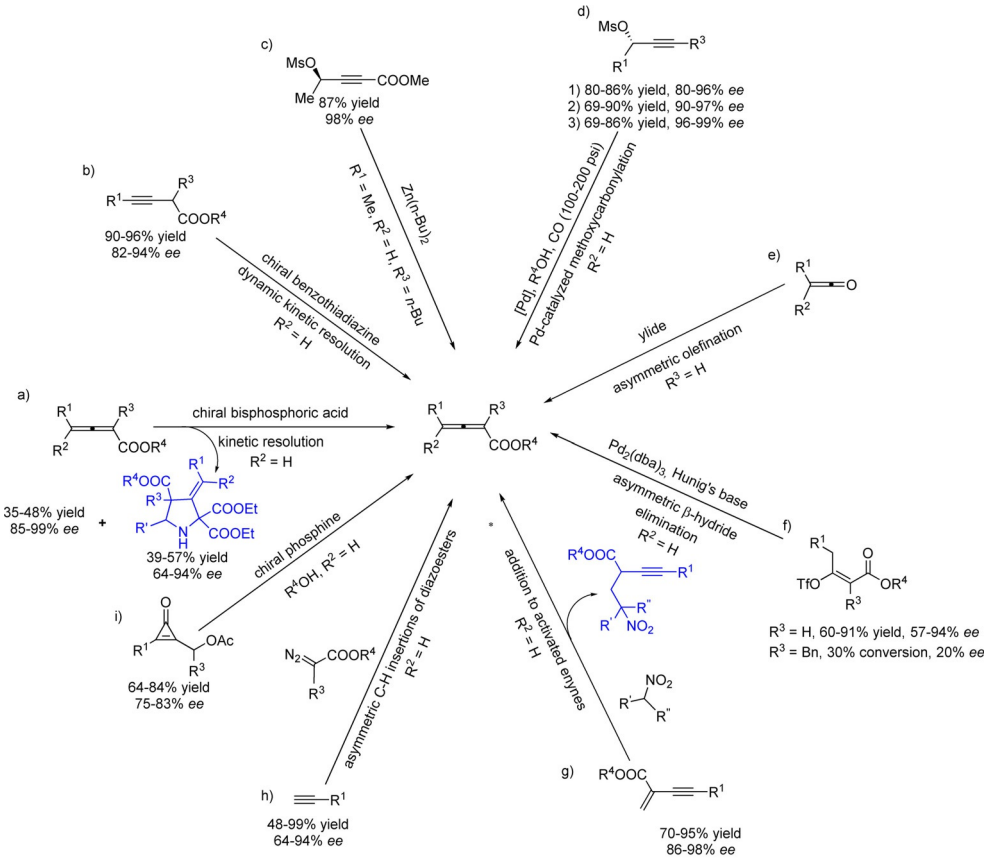
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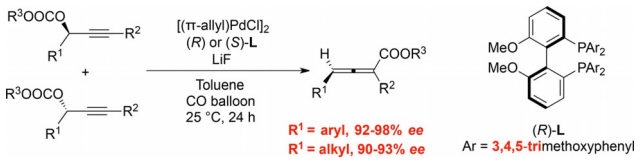
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Scheme 1. Reported methods for the synthesis of chiral trisubstituted 2,3-allenoates.



Scheme 2. Enantioselective synthesis of 2,3-allenoates from racemic propargylic carbonates.

catalyst at room-temperature to investigate the ligand effect for this methoxycarbonylation reaction. Our goal was to finely tune the electronic property of the phosphorus atoms in the BIPHEP skeleton: Firstly, changing the aryl substituents of the coordinating phosphorus atom from 3,5-dimethoxy to 3,5-dimethyl [(*R*)-**L1**] led to 90% recovery of the starting material and (R_a)-**2a** was detected with only 4% yield (Table 1, entry 1). Replacing the aryl substituents with 4-methylphenyl to decrease the electron density of the phosphorous atom [(*R*)-**L2**] led to inferior results (Table 1, entry 2). To our disappointment, when more electron-rich ligands (*R*)-3,5-*i*Pr$_2$-4-Me$_2$N-MeOBIPHEP [(*R*)-**L3**] and (*R*)-Ph-Garphos [(*R*)-**L4**] were surveyed, the carbonate (±)-**1a** remained mostly unreacted and only a trace amount of the target product was detected (Table 1, entries 3 and 4). Interestingly, when 3,4,5-trimethoxy groups were introduced [(*R*)-**L5**], and after stirring at room temperature for 24 h, (R_a)-**2a** was obtained in 55% yield with 98% *ee* (Table 1, entry 6). Inspired by this observation, the catalyst loading was increased in an attempt to raise the conversion of this methoxycarbonylation process. To our delight, when 3 mol% [(π-allyl)PdCl]$_2$ was used in combination with 9 mol% (*R*)-**L5**, the desired product (R_a)-**2a** was obtained in 85% yield and 97% *ee* with the starting material being completely converted (Table 1, entry 7). By comparison, when (*R*)-**ECNU-phos** was used, (R_a)-**2a** was produced in 78% yield with only 92% *ee* (Table 1, entry 5).[15]

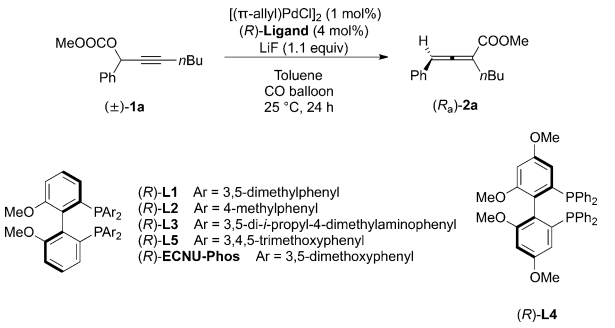
Having identified the optimal chiral ligand, other experimental parameters were inspected to further optimize the reaction conditions. Firstly, [(π-allyl)PdCl]$_2$ was replaced by [(π-cinnamyl)PdCl]$_2$ and the product was obtained with 58% yield and 97% *ee* (Table 2, entry 2). Considering its lower cost, [(π-allyl)PdCl]$_2$ was used for further study.[16] Among the solvents surveyed, toluene was shown to be the best (Table 2, entries 3–8). 1,4-Dioxane was also effective, albeit with a decreased yield and lower enantioselectivity (Table 2, entry 3). The enatioselectivity and yield dropped further when THF was used (Table 2, entry 4) and the methoxycarbonylation proceeded slowly in DCM to afford only 19% of (R_a)-**2a** after 24 h of stirring (Table 2, entry 5). When MTBE, DMF, or CH_3CN was applied, the catalyst system was ineffective with only a trace amount of target products being detected (Table 2, entries 6–8).

After further optimization, it was apparent that when 3 mol% [(π-allyl)PdCl]$_2$ and 9 mol% (*R*)-**L5** were applied as the chiral catalyst in toluene under 1 atm of CO with 1.1 eqiuv LiF

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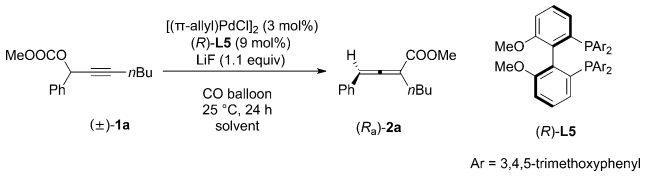
Table 1. Pd-catalyzed enantioselective methoxycarbonylation of propargylic carbonates (±)-**1 a**-ligand effect.[a]



Entry	Ligand	(R_a)-**2 a** Yield [%]	*ee* [%]	Recovery of (±)-**1 a** [%][b]
1	(*R*)-**L1**	4[b]	–	90
2	(*R*)-**L2**	6[b]	–	87
3	(*R*)-**L3**	6[b]	–	80
4	(*R*)-**L4**	4[b]	–	92
5[c]	(*R*)-**ECNU-Phos**	78[d]	92[e]	21
6[f]	(*R*)-**L5**	55[d]	98[e]	28
7[c]	(*R*)-**L5**	85[d]	97[e]	N.D.

[a] Reaction conditions: (±)-**1 a** (0.2 mmol), [(π-allyl)PdCl]$_2$ (1 mol%), (*R*)-**L** (4 mol%) and LiF (1.1 equiv) at 25 °C in toluene (2.0 mL) under 1 atm of CO unless otherwise noted. [b] Determined by ^{1}H NMR analysis using 1,3,5-trimethylbenzene as the internal standard; N.D. = not detected. [c] 0.1 mmol (±)-**1 a** was used in combination with 3 mol% [(π-allyl)PdCl]$_2$ and 9 mol% (*R*)-**L**. [d] Isolated yields. [e] *ee* values of 2,3-allenoates were determined by HPLC. [f] The reaction was carried out with 0.1 mmol (±)-**1 a**.

Table 2. Optimization of reaction conditions.[a]

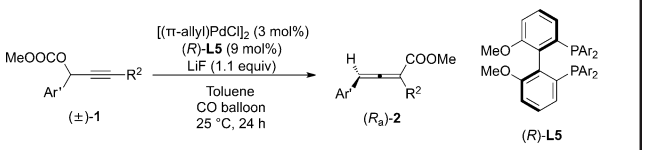


Entry	Solvent	(R_a)-**2 a** Yield [%][b]	*ee* [%][c]	Recovery of (±)-**1 a** [%][d]
1	Toluene	85	97	N.D.
2[e]	Toluene	58	97	20
3	1,4-Dioxane	70	95	N.D.
4	THF	50	92	N.D.
5	DCM	19[d]	–	56
6	MTBE	N.D.	–	77
7	DMF	5[d]	–	N.D.
8	CH_3CN	5[d]	–	81

[a] Reaction conditions: (±)-**1 a** (0.1 mmol), [(π-allyl)PdCl]$_2$ (3 mol%), (*R*)-**L5** (9 mol%) and LiF (1.1 equiv) at 25 °C in toluene (2.0 mL) under 1 atm of CO unless otherwise noted; MTBE = methyl *tert*-butyl ether. [b] Isolated yield. [c] *ee* values of 2,3-allenoates were determined by HPLC. [d] Determined by ^{1}H NMR analysis using 1,3,5-trimethylbenzene as the internal standard; N.D = not detected. [e] 0.2 mmol (±)-**1 a**, 1 mol% [(π- cinnamyl)PdCl]$_2$, and 4 mol% (*R*)-**L5** were used.

as additive, 2,3-allenoates (R_a)-**2** were obtained in the highest yield and enantiopurity after 24 h of stirring at room temperature (25 °C). With the optimized conditions in hand, the substrate scope was then investigated: When 1-aryl propargylic carbonates were used, (*R*)-**L5** exhibited excellent activity, producing (R_a)-**2** with very high enatioselectivities and recovery of starting materials were not detected. When the simple phenyl-substituted substrate (±)-**1 a** was applied, (*R*)-**L5** resulted in 86% yield and 97% *ee* on 1 mmol scale (Table 3, entry 1 a). Different substituents could be introduced onto the Ar′ group; both electron-donating and -withdrawing groups were well tolerated, affording the corresponding allenonates in good yields with excellent enatioselectivities (Table 3, entries 2, 3 a, 4, 5). Aryl groups with *m*-Cl and *p*-Cl made no difference, furnishing the target products (R_a)-**2 f** and (R_a)-**2 g** with 82–84% yields and 97–98% *ee* (Table 3, entries 6 and 7 a). The reactions of naphthyl-substituted substrates **1 h** and **1 i** were easily carried out to afford (R_a)-**2 h** and (R_a)-**2 i** with 80% yield and excellent enatioselectivities (Table 3, entries 8 and 9). The R^2 group is also very flexible: methyl, isopropyl, long-chain *n*-hexadecyl, phenethyl, 4-chlorobutyl, and cyclohexyl all performed very well in this reaction, affording the products with good yields and no less than 92% *ee* (Table 3, entries 10–12, 13 a, 14–16). Compared to (*R*)-**ECNU-Phos**, an impressive enantioselectivity was observed (Table 3, entries 1 b, 3 b, 7 b, and 13 b). The reac-

Table 3. Substrate scope of enantioenriched allenoates synthesis from racemic 1-aryl propargylic carbonates (±)-1.[a]



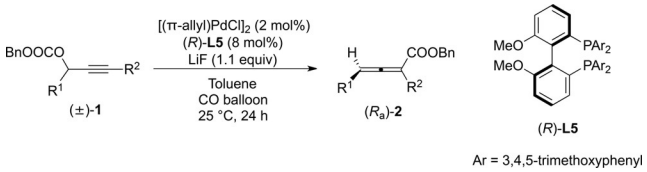
Entry	(±)-**1** Ar′	R^2	Yield of (R_a)-**2** [%][b]	*ee* of (R_a)-**2** [%][c]
1 a	Ph	*n*Bu (**1 a**)	86 (**2 a**)	97
1 b[d]	Ph	*n*Bu (**1 a**)	77 (**2 a**)	93
2	4-MeC_6H_4	*n*Bu (**1 b**)	80 (**2 b**)	98
3 a	3-MeOC_6H_4	*n*Pr (**1 c**)	88 (**2 c**)	97
3 b[e]	3-MeOC_6H_4	*n*Pr (**1 c**)	85 (**2 c**)	92
4[f]	4-NCC_6H_4	*n*Bu (**1 d**)	72 (**2 d**)	95
5[f]	4-$MeO_2CC_6H_4$	*n*Bu (**1 e**)	79 (**2 e**)	96
6	3-ClC_6H_4	*n*Bu (**1 f**)	84 (**2 f**)	97
7 a	4-ClC_6H_4	*n*Bu (**1 g**)	82 (**2 g**)	98
7 b[g]	4-ClC_6H_4	*n*Bu (**1 g**)	87 (**2 g**)	91
8[h]	1-naphthyl	*n*Bu (**1 h**)	80 (**2 h**)	95
9[h]	2-naphthyl	*n*C_8H_{17} (**1 i**)	80 (**2 i**)	93
10	Ph	Me (**1 j**)	85 (**2 j**)	92
11	Ph	*i*Pr (**1 k**)	84 (**2 k**)	98
12	Ph	*n*$C_{16}H_{33}$ (**1 l**)	86 (**2 l**)	97
13 a	Ph	$(CH_2)_2$Ph (**1 m**)	81 (**2 m**)	97
13 b[i]	Ph	$(CH_2)_2$Ph (**1 m**)	79 (**2 m**)	93
14	Ph	$(CH_2)_4$Cl (**1 n**)	83 (**2 n**)	97
15	Ph	allyl (**1 o**)	85 (**2 o**)[g]	95
16	4-EtC_6H_4	Cy (**1 p**)	83 (**2 p**)	97

[a] Reaction conditions: (±)-**1** (1.0 mmol), [(π-allyl)PdCl]$_2$ (3 mol%), (*R*)-**L5** (9 mol%), and LiF (1.1 equiv) at 25 °C in toluene (5.0 mL) under 1 atm of CO unless otherwise noted. [b] Isolated yield. [c] *ee* values of 2,3-allenoates were determined by HPLC. [d] Using (*R*)-**ECNU-phos**; 17% of **1 a** was recovered.[15] [e] With (*R*)-**ECNU-phos**; 6% of **1 c** was recovered.[15] [f] The reaction was carried out in 35 °C. [g] Using (*R*)-**ECNU-phos**.[15] [h] The reaction was carried out for 26 h. [i] Using (*R*)-**ECNU-phos**; 10% of **1 m** was recovered.[15]

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Table 4. Substrate scope of enantioenriched allenoates synthesis from racemic 1-alkyl-substituted propargylic carbonates (±)-**1**.[a]

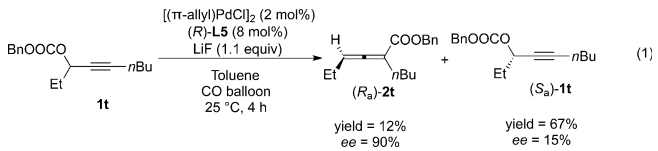


Entry	(±)-**1** R^1	R^2	Yield of (R_a)-**2** [%][b]	*ee* of (R_a)-**2**[%][c]	Recovery of (±)-**1**[%][d]
1	Me	*i*Pr (**1q**)	23 (**2q**)	92	66
2	Me	*n*Bu (**1r**)	31 (**2r**)	91	54
3	Me	C_6H_{13} (**1s**)	24 (**2s**)	92	61
4	Et	*n*Bu (**1t**)	38 (**2t**)	90	49
5a	Et	nC_8H_{17} (**1u**)	38 (**2u**)	90	60
5b[e]	Et	nC_8H_{17} (**1u**)	52 (**2u**)	85	26
6a[f]	*n*Bu	*n*Pr (**1v**)	25 (**2v**)	92	66
6b[e]	*n*Bu	*n*Pr (**1v**)	51 (**2v**)	84	21
7a[f]	*n*Bu	nC_7H_{15} (**1w**)	27 (**2w**)	90	67
7b[e]	*n*Bu	nC_7H_{15} (**1w**)	52 (**2w**)	84	18
8a[f]	nC_6H_{13}	*n*Bu (**1x**)	24 (**2x**)	91	65
8b[e]	nC_6H_{13}	*n*Bu (**1x**)	54 (**2x**)	83	15
9a[f]	nC_6H_{13}	C_8H_{17} (**1y**)	27 (**2y**)	93	58
9b[e]	nC_6H_{13}	C_8H_{17} (**1y**)	50 (**2y**)	84	22
10a[f]	$nC_{11}H_{23}$	*n*Pr (**1z**)	35 (**2z**)	90	56
10b[e]	$nC_{11}H_{23}$	*n*Pr (**1z**)	54 (**2z**)	84	15

[a] Reaction conditions : (±)-**1** (1.0 mmol), [(π-allyl)PdCl]$_2$ (2 mol%), (*R*)-**L5** (8 mol%) and LiF (1.1 equiv) at 25 °C in toluene (5.0 mL) under 1 atm of CO unless otherwise noted. [b] Isolated yield. [c] *ee* values of 2,3-allenoates were determined by HPLC. [d] Determined by ^{1}H NMR analysis using 1,3,5-trimethylbenzene as the internal standard. [e] The reaction was carried out with (*R*)-**ECNU-phos**. [f] The reactions were carried out at 20 °C, data for reactions carried out at 25 °C are available in the Supporting Information.

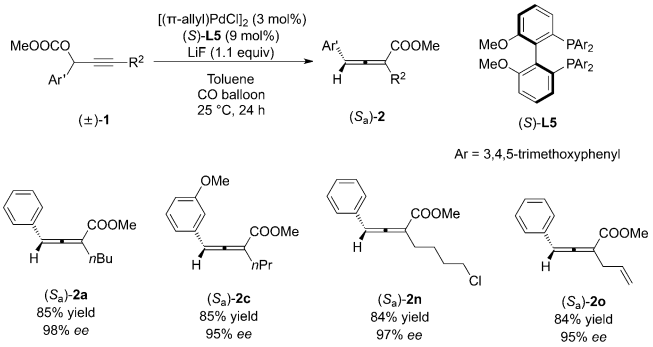
tion of the substrate in which Ar′ and R^2 were both phenyl was very slow.

Furthermore when the R^1 group was methyl, allenoates **2q–s** could be accessed with *ee* values higher than 91% using (*R*)-**L5** (Table 4, entries 1–3). Changing R^1 group from methyl to ethyl, (R_a)-**2t** and (R_a)-**2u** were provided in 90% *ee* using (*R*)-**L5** (Table 4, entries 4, 5a). Further lengthening of the R^1 group to *n*-butyl, *n*-hexyl, and *n*-undecyl did not impact the enantioselectivity using (*R*)-**L5**, which afforded allenoates with 90–93% *ee*. By contrast, (*R*)-**ECNU-phos** resulted in these allenoates with only 83–85% *ee* (Table 4, entries 5b, 6b, 7b, 8b, 9b, and 10b). The lower yielding nature for substrates in which R^1 was an alkyl substituent may be caused by the difficulty in removing leaving groups as compared to the reaction of substrates with R^1 being an aryl group, in which the carbonate group is placed at the benzylic position. In addition, it should be noted that the *ee* value of the recovered starting material was 15%, indicating that the reaction is at least practically a kinetic resolution process, as shown in Equation (1):



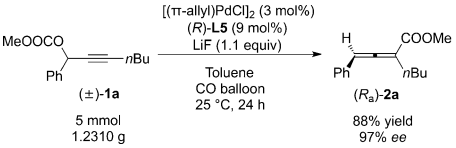
(1)

(*S*)-**L5** was also effective in this methoxycarbonylation process, affording the *S*-products (Scheme 3). When the Ar′ group is phenyl and substituted with electron-donating groups, the enantiomers could be obtained with similar yields and *ee* values. The chlorine-substituted substrate [(±)-**1n**] could be transformed into the desired enantiomer with 97% *ee*. It is worth mentioning that an allyl group can also be accommodated, affording (S_a)-**2o** with 84% yield and 95% *ee*.



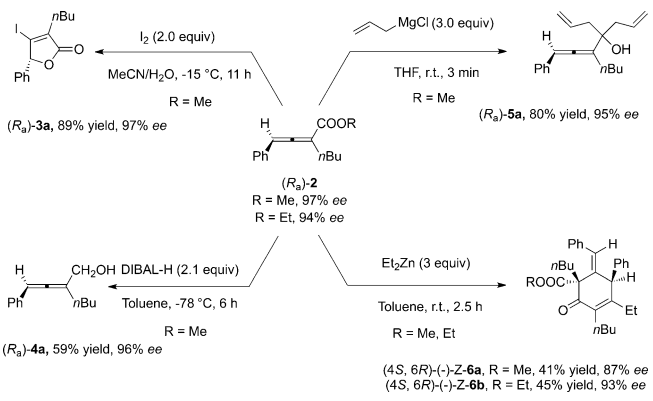
Scheme 3. (*S*)-3,4,5-(MeO)$_3$-MeOBIPHEP accelerated Pd-catalyzed methoxycarbonylation.

The practicality was demonstrated by subjecting 5 mmol of (±)-**1a** to the reaction, which afforded (R_a)-**2a** in 88% yield and 97% *ee* under the standard catalytic conditions, as shown in Equation (2):



(2)

As a class of very useful compounds in organic synthesis, the 2,3-allenoates may be easily transformed into other versatile chemicals with high *ee* values (Scheme 4). Treatment with I_2 at −15 °C, the lactone (R_a)-**3a** could be afforded in 89% yield with 97% *ee*. In the presence of DIBAL-H, (R_a)-**2a** may be transferred into the trisubstituted 2,3-allenol (R_a)-**4a** with 96% *ee*,



Scheme 4. Synthetic applications of 2,3-allenoate (R_a)-**2**.

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which is a versatile precursor for halides, mesylates, tosylates, amines, thiols, etc.[17] (R_a)-**2a** could also undergo a 1,2-addition reaction with allyl magnesium chloride,[18] furnishing the chiral tertiary α-allenol (R_a)-**5a** in 80% yield and 95% *ee*. The reaction of 2,3-allenoates with organozinc compounds is an efficient way to access 5-benzylidenecyclohex-2-enones through a double Michael addition/cyclization.[19] Thus, after stirring for 2.5 h at room temperature, (4*S*,6*R*)-(−)-*Z*-**6a** could be afforded with 87% *ee*. Interestingly, utilizing the ethyl ester led to a much higher chirality transfer efficiency, furnishing (4*S*,6*R*)-(−)-*Z*-**6b** with 93% *ee*.

In summary, we have succeeded in establishing a chiral catalyst system consisting of $[(\pi\text{-allyl})PdCl]_2$ and (*R*) or (*S*)-3,4,5-$(MeO)_3$-MeOBIPHEP that facilitates the Pd-catalyzed highly enantioselective alkoxycarbonylation of readily available racemic propargylic carbonates, affording optically active 2,3-allenoates with 90–98% *ee*. Further studies on this topic are being conducted in our laboratory.

Experimental Section

To a flame-dried Schlenk flask (100 mL) was added $[(\pi\text{-allyl})PdCl]_2$ (56.1 mg, 0.15 mmol) and (*R*)-**L5** (438.2 mg, 0.45 mmol). The flask was then degassed and refilled with argon three times to ensure the complete exclusion of air. Then, freshly distilled toluene (15 mL) was added under argon. The resulting mixture was stirred for 1 h at room temperature, followed by the addition of anhydrous LiF (143.3 mg, 5.5 mmol; kept in a glove box), (±)-**1a** (1.2310 g, 5.0 mmol), and toluene (10 mL) sequentially. The mixture was then frozen with a liquid nitrogen bath, degassed to remove the argon inside completely, and refilled with CO three times using a balloon of CO. Then the solution was stirred at 25 °C for 24 h. After that, the mixture was diluted with Et_2O (25 mL), washed with brine (20 mL), and dried over anhydrous Na_2SO_4. The resulting mixture was filtered through a short silica gel column (5.0 cm) eluted with Et_2O (50 mL), and the resulting eluent concentrated. The residue was purified by column chromatography on silica gel to afford (R_a)-**2a**[15] (1.0129 g, 88%) as an oil [eluent: petroleum ether (b.p. 60–90 °C)/ethyl ether = 250/1] in 97% *ee* (HPLC conditions: IC column, hexane/*i*PrOH = 200/1, 0.5 mL min^{-1}, λ = 214 nm, t_R (minor) = 5.4 min, t_R (major) = 6.9 min); $[\alpha]^{23}_D = -47.4$ (c = 1.04, $CHCl_3$) [93% *ee*, $[\alpha]^{21}_D = -47.3$ (c = 1.15, $CHCl_3$)][15]; ^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.38–7.22 (m, 5H, Ar-H), 6.53 (t, J = 2.8 Hz, 1H, =CH), 3.74 (s, 3H, CH_3), 2.45–2.29 (m, 2H, CH_2), 1.53–1.32 (m, 4H, 2 *x* CH_2), 0.89 ppm (t, J = 7.4 Hz, 3H, CH_3); ^{13}C NMR (100 MHz, $CDCl_3$): δ = 212.1, 167.2, 132.5, 128.8, 127.7, 127.2, 104.3, 98.3, 52.2, 30.2, 28.6, 22.3, 13.8 ppm.

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Conflict of interest

The authors declare no conflict of interest.

Keywords: alkoxycarbonylation · allenoates · enatioselective catalysis · palladium · propargylic carbonates

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